Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

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Colour code used in these notes and in the slides
current i voltage v a part of a circuit (box N), terminals a, b

Motivation. So far, KCL and KVL were written for any circuit, with any elements. From now on we restrict the elements. In an **LTI resistive circuit** every equation is linear and algebraic. Such a circuit can be solved in two ways: by transforming it into a simpler circuit, or by writing the equations of the circuit as it is. This lecture gives the tools for the first way: the substitution theorem, one-ports and their equivalents, series and parallel connections, and the exchange of a voltage source and a current source.

1. LTI resistive circuits

First classify the elements, then the circuit. Independent sources are left aside: they are the inputs (see below).

Resistive and dynamic elements

- A **resistive** element relates its voltages and currents at the same instant by an algebraic law; it has no memory. Examples: resistors, diodes, dependent sources, ideal transformers, and the ideal op-amp met later. "Resistive" covers all of them, not only resistors.
- A **dynamic** element has a derivative in its law, so it has memory: capacitors ($i = C\,dv/dt$), inductors, coupled inductors.
- An element is **linear** if its law is linear in its voltages and currents, and **time-invariant** if its law does not change with time.

Nonlinear, time-varying, dynamic circuits

- A **nonlinear** circuit contains at least one nonlinear element, e.g. a diode. Otherwise it is **linear**.
- A **time-varying** circuit contains at least one time-varying element, e.g. $R(t)$. Otherwise it is **time-invariant**.
- A **dynamic** circuit contains at least one dynamic element: an L or a C. Otherwise it is **resistive**.

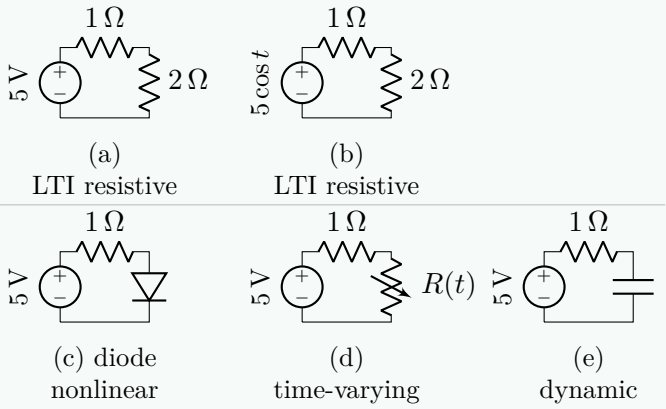
An **LTI resistive circuit** is linear, time-invariant and resistive: apart from the independent sources, every element is linear, time-invariant and resistive (LTI resistors, linear dependent sources, ideal transformers, ...).

Independent sources are inputs

An independent source is an **external input**: a given signal of time, such as $v_s(t) = 5\cos t$ V. The three pairs of words describe the circuit that the inputs drive, not the inputs.

What varies in time? In every circuit the voltages and currents are functions of time. What matters is the *element law*. A time-invariant resistor obeys $v(t) = R\,i(t)$ with the same R at every instant; a time-varying one obeys $v(t) = R(t)\,i(t)$: the law itself changes. Linear and nonlinear also refer to the element law.

Example 10.1 — which circuit is which?



Each circuit is a source, a $1\,\Omega$ resistor and one more element.

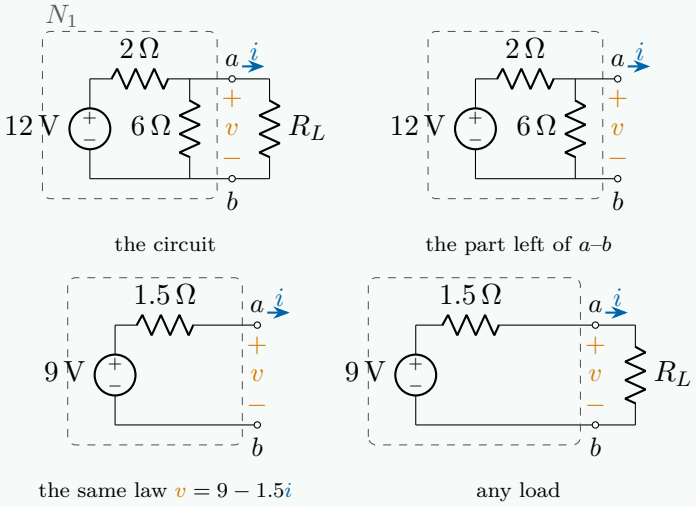
Top row: the same circuit with two inputs. Both are LTI resistive: the input does not change the class.

Bottom row: the same input, the $2\,\Omega$ replaced. Each of (c), (d), (e) breaks exactly one property. In (d), $R(t) = 2 + \sin t\ \Omega$.

What we gain. KCL and KVL are *always* linear: they come from the graph and need no assumption on the elements. In an LTI resistive circuit every element equation is linear and algebraic too: $v = Ri$ for a resistor, $v = \mu v_x$ or $i = g\,v_x$ for a dependent source, $v_2 = nv_1$ and $i_1 = -ni_2$ for an ideal transformer. No derivative appears. Time enters only through the values of the independent sources, so at each instant the circuit is solved like a circuit with constant sources.

2. A first example

Example 10.2 — what does R_L see?



The load R_L can be any resistive element. Find v and i.

Leave R_L aside and look only at the part left of a–b. The $2\,\Omega$ has $12 - v$ across it, the $6\,\Omega$ has v. KCL at a:

$$\frac{12 - v}{2} = \frac{v}{6} + i \quad \Rightarrow \quad v = 9 - 1.5\,i.$$

A 9 V source in series with $1.5\,\Omega$ obeys the same law (KVL), so R_L cannot tell the two apart.

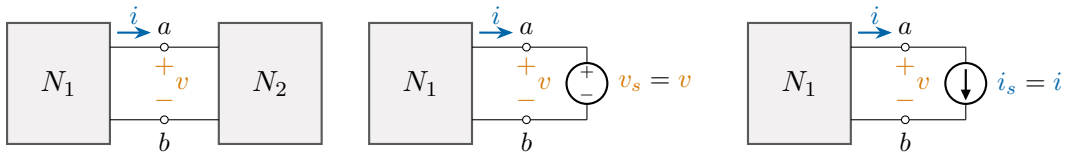
Any load. With $v = R_L i$: $i = 9/(1.5 + R_L)$. For $R_L = 3\,\Omega$: $i = 2\,\text{A}$, $v = 6\,\text{V}$.

What did we use? The part left of a–b became one source and one resistor, and R_L did not notice. The work is done once; each new load is one line of algebra. Three questions make this precise:

- May a part be replaced by a source? — the **substitution theorem** (Section 3).
- When may a part be replaced by a simpler part? — **equivalent one-ports** (Section 4).
- Which simple replacements are there? — **series, parallel**, **source transformation** (Sections 5, 6).

3. The substitution theorem

Split a circuit into two parts N_1 and N_2, joined at two terminals a and b. Let $v(t)$ be the voltage between a and b, and $i(t)$ the current from N_1 into N_2 at a.



(I) the circuit (II) $N_2 \to$ voltage source (III) $N_2 \to$ current source

Substitution theorem

Replace N_2 by an independent voltage source $v_s(t) = v(t)$, or by an independent current source $i_s(t) = i(t)$. No branch voltage or branch current inside N_1 changes. An observer inside N_1 cannot tell (I), (II) and (III) apart. Condition: each circuit has a unique solution.

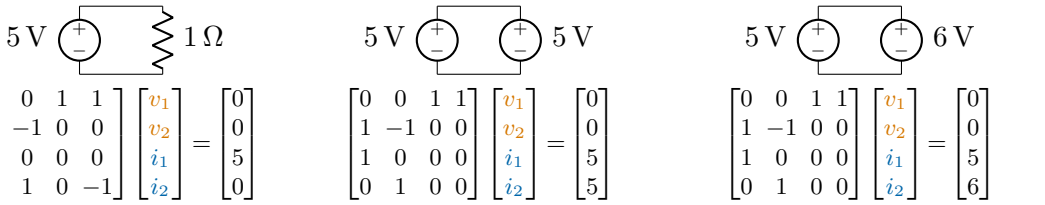
Unique solution

A circuit with n nodes and b branches has $2b$ unknowns, the branch voltages $\mathbf{v}$ and currents $\mathbf{i}$, and $2b$ equations:

KCL	$A\,\mathbf{i} = 0$	$n-1$
KVL	$B\,\mathbf{v} = 0$	$b-n+1$
element laws	one per branch	b

The circuit has a **unique solution** if exactly one $(\mathbf{v}, \mathbf{i})$ satisfies all $2b$ equations.

In matrix form. Collect the $2b$ unknowns in one vector $x = (v_1, \ldots, v_b, i_1, \ldots, i_b)$. In an LTI resistive circuit all $2b$ equations are linear, so they stack into $Tx = u$. Here T, the *coefficient matrix*, has one row per equation and holds the numbers that multiply the unknowns; u holds the source values. The solution is unique exactly when T is invertible, $\det T \neq 0$. Below, branch 1 is the left element, branch 2 the right one, both oriented downwards.



$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \end{bmatrix}$$

(a) $\det T = -1$: one solution

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 5 \end{bmatrix}$$

(b) $\det T = 0$: many

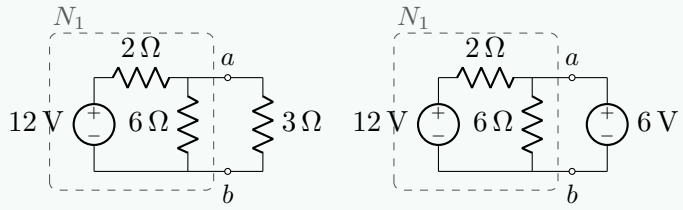
$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 6 \end{bmatrix}$$

(c) $\det T = 0$: none (KVL fails)

In (b) and (c), i_1 and i_2 appear only in the KCL row, so T is not invertible.

(I) and (II): replacing N_2 by a voltage source

Example 10.3 — a voltage source in the first example

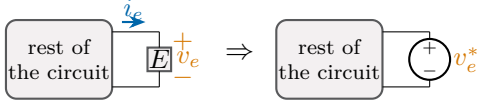


(I) N_2: $R_L = 3\,\Omega$ (II) a 6 V source

(I) From Example 10.2 with $R_L = 3\,\Omega$: $v = 6\,\text{V}$, $i = 2\,\text{A}$; the $2\,\Omega$ carries 3 A, the $6\,\Omega$ carries 1 A.

(II) KCL at a with $v = 6\,\text{V}$: $(12-6)/2 = 6/6 + i$, so $i = 2\,\text{A}$. Again 3 A and 1 A: nothing changed. In (I) we solved the same equation, since there too $v = 6\,\text{V}$.

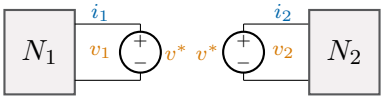
Why it works, step 1: one element. Take any element E of a circuit, with v_e^* and i_e^* in the solution, and replace only E by a voltage source v_e^*. Assume both circuits have a unique solution.



	before	after
KCL, KVL	$A\mathbf{i} = 0$, $B\mathbf{v} = 0$	same graph: the same
other $b-1$ elements	their laws	the same
element E	the law of E	$v_e = v_e^*$

The old solution $(\mathbf{v}^*, \mathbf{i}^*)$ satisfies all old equations, the law of E included. So it satisfies the unchanged rows of the new circuit, and $v_e = v_e^*$ holds too; a voltage source takes any current, so $i_e = i_e^*$ is allowed. With a unique solution, nothing changes. Uniqueness of the old circuit alone is not enough: in (a), replacing the $1\,\Omega$ by its voltage $5\,\mathrm{V}$ gives (b).

Step 2: divide N_1 and N_2. Let v_1, i_1 be the port values of N_1 and v_2, i_2 those of N_2. Divide the two parts and give each its own voltage source v^*. Assume a unique solution.

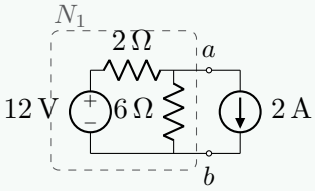


	(I)	divided
inside N_1, inside N_2	KCL, KVL, element laws	the same
the connection	$v_1 = v_2$, $i_1 = i_2$	$v_1 = v_2 = v^*$

The solution of (I) satisfies these equations, since $v_1 = v_2 = v^*$ there. With a unique solution, nothing changes in either part. The left part is exactly (II), so nothing inside N_1 changes, also $i_1 = i^*$. N_2 is not needed at all: it can be anything, even nonlinear.

(I) and (III): replacing N_2 by a current source

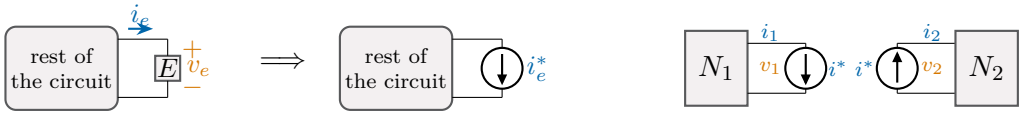
Example 10.4 — a current source in the first example



(III) a 2 A source

KCL at a with $i = 2\,\mathrm{A}$: $(12 - v)/2 = v/6 + 2$, so $v = 6\,\mathrm{V}$. Again the $2\,\Omega$ carries $3\,\mathrm{A}$ and the $6\,\Omega$ carries $1\,\mathrm{A}$: nothing changed. The same equation holds in (I), where $i = 2\,\mathrm{A}$.

Why it works. The same two steps, now with current sources.

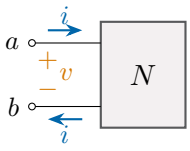


Step 1: replacing E by a current source i_e^* changes only its law, to $i_e = i_e^*$, which the old solution satisfies; a current source takes any voltage. Step 2: the connection becomes $i_1 = i_2 = i^*$. So $v_1 = v^*$. Uniqueness is needed: two $2\,\mathrm{A}$ sources in series have many solutions.

4. Ports and equivalent one-ports

Port, one-port

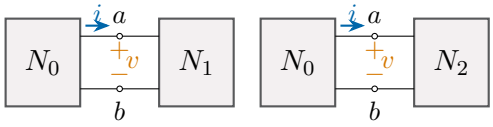
A pair of terminals is a **port** if the current entering one terminal equals the current leaving the other. A **one-port** is a circuit, or a circuit element, with one pair of terminals accessible from outside.



A one-port is always a port: by KCL on a closed surface around N, the current i that enters at a leaves at b. We take i entering the $+$ terminal of v. The relation between v and i is the i–v **characteristic** of the one-port. A resistor is the simplest one-port: $v = Ri$.

Equivalent one-ports

Two one-ports N_1 and N_2 are **equivalent**, $N_1 \equiv N_2$, if they have the same i–v characteristic at their terminals. Then, for any one-port N_0 connected to them, every branch voltage and branch current inside N_0 is the same with N_1 or with N_2.

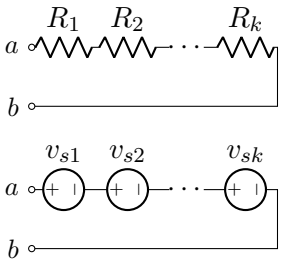


N_0 meets the rest only through v and i. The equations of the whole circuit are those of N_0 plus the i–v characteristic at the port. Equivalent one-ports give the same equations, so the same solution inside N_0. Inside N_1 and N_2 the branch values may differ.

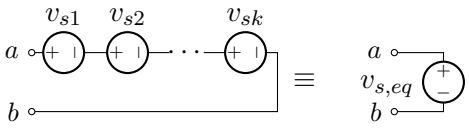
5. Series and parallel connections

Series connection

In a series connection the same current i flows through every element (KCL), and the voltages add (KVL).



$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

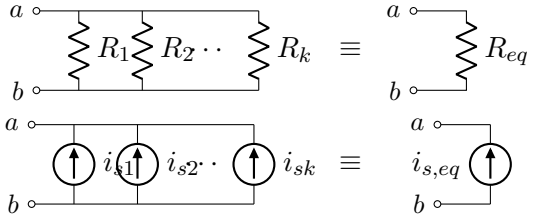


$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

For the resistors: $v = R_1 i + \cdots + R_k i = R_{eq} i$. The share of R_j is $v_j = (R_j/R_{eq})\, v$ (**voltage divider**). For the sources all $+$ signs point towards a; a source turned the other way enters the sum with $-$.

Parallel connection

In a parallel connection every element has the same voltage v (KVL), and the currents add (KCL).



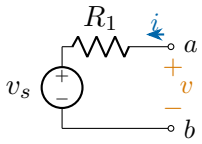
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

$$i_{s,eq} = i_{s1} + i_{s2} + \cdots + i_{sk}$$

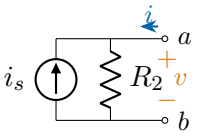
With conductances $G_j = 1/R_j$: $i = G_1 v + \cdots + G_k v = G_{eq} v$, so $G_{eq} = G_1 + \cdots + G_k$. The share of R_j is $i_j = (G_j/G_{eq})\, i$ (**current divider**). Two resistors: $R_{eq} = R_1 R_2/(R_1 + R_2)$.

6. One source and one resistor

Take a voltage source with a series resistor, and a current source with a parallel resistor. In both, i enters at a.



$$v = R_1 i + v_s \quad (1)$$



$$v = R_2(i + i_s) = R_2 i + R_2 i_s \quad (2)$$

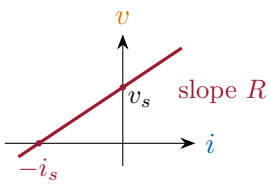
In (2), i and i_s both flow into the top node, so R_2 carries $i + i_s$. Both characteristics are straight lines in the i–v plane. They are the same line when the slopes and the intercepts agree.

Source transformation

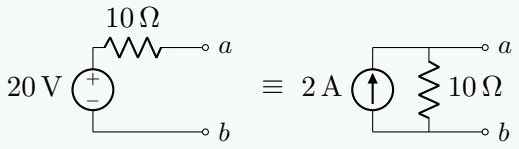
The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$. Short circuit ($v = 0$): $i = -i_s$.

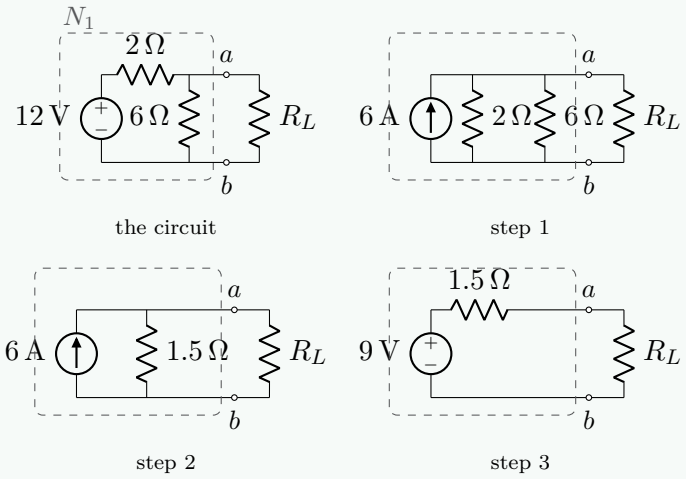


Example 10.5 — a source transformation



$R = 10\,\Omega$ and $i_s = v_s/R = 20/10 = 2\,\text{A}$. Both have open-circuit voltage 20 V and short-circuit current 2 A (out of a).

Example 10.6 — Example 10.2 again, by source transformations



Only the part left of a–b is simplified; R_L stays.

1. Source transformation: 12 V with 2 Ω becomes 6 A with 2 Ω ($12 = 2 \times 6$).

2. Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$.

3. Source transformation: 6 A with 1.5 Ω becomes 9 V with 1.5 Ω ($9 = 1.5 \times 6$).

The result is the same 9 V and 1.5 Ω. Inside, the values differ: for $R_L = 3\,\Omega$ the 12 V source carries 3 A, the 9 V source 2 A.

7. Solving LTI resistive circuits

Series and parallel connections and source transformations simplify many circuits, but not all. Two methods write the equations of the circuit as it is, without transforming it, and work for every LTI resistive circuit:

1. **node voltage analysis** — the $n-1$ node voltages are the unknowns;
2. **mesh current analysis** — the $b-n+1$ mesh currents are the unknowns.

Both start from the matrices met earlier (A, M), now with linear element equations.

Summary

- A **resistive** element has an algebraic law with no memory; a **dynamic** one (L, C) has a derivative. In an LTI resistive circuit every element, apart from the independent sources (inputs), is linear, time-invariant and resistive.
- **Substitution:** a part may be replaced by a source equal to its port voltage or current; nothing changes in the rest (if each circuit has a unique solution).
- Two one-ports are **equivalent** if they have the same i–v characteristic; the rest of the circuit cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$. Source transformation: v_s with series $R \equiv i_s = v_s/R$ with parallel R.